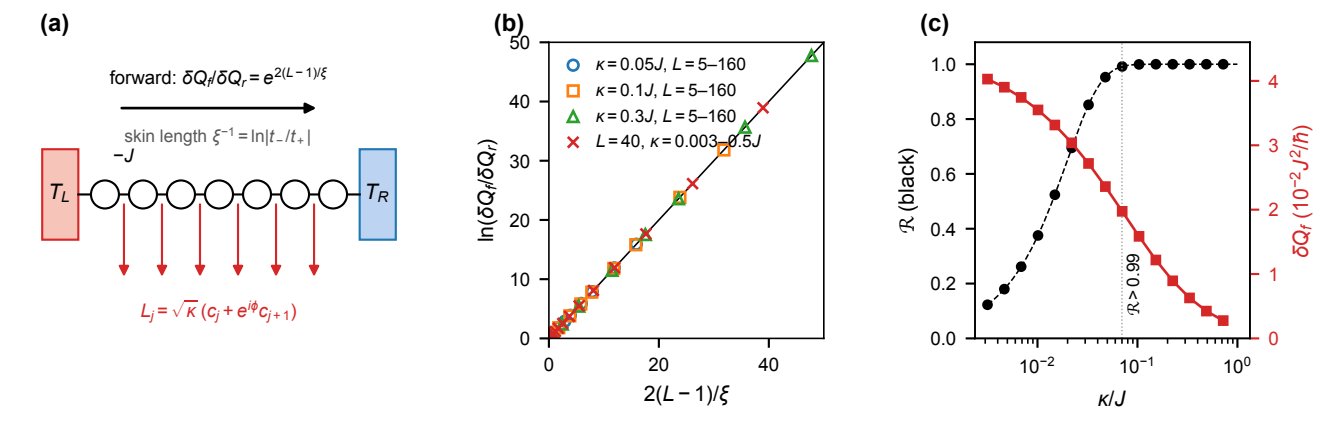

(a)
forward: $\delta Q_l/\delta Q_r = e^{2(L-1)/\xi}$
skin length $\xi^{-1} = \ln|t_-/t_+|$
$-J$
T_L
T_R
$L_j = \sqrt{\kappa}\,(c_j + e^{i\phi} c_{j+1})$
(b)
$\kappa = 0.05J$, $L = 5$–160
$\kappa = 0.1J$, $L = 5$–160
$\kappa = 0.3J$, $L = 5$–160
$L = 40$, $\kappa = 0.003$–$0.5J$
$\ln(\delta Q_l/\delta Q_r)$
$2(L-1)/\xi$
(c)
$\mathcal{R}$ (black)
$\mathcal{R} > 0.99$
δQ_r ($10^{-2} J^2/\hbar$)
κ/J